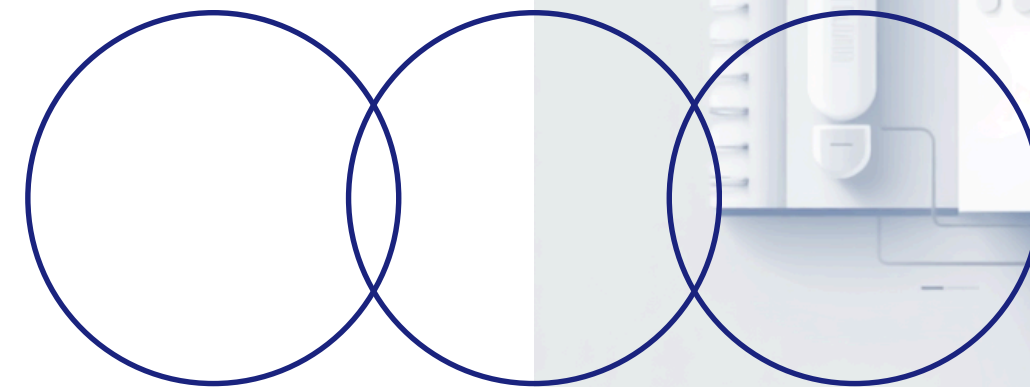


HPC INNOVATIONS

SLURM Magic in Jupyter

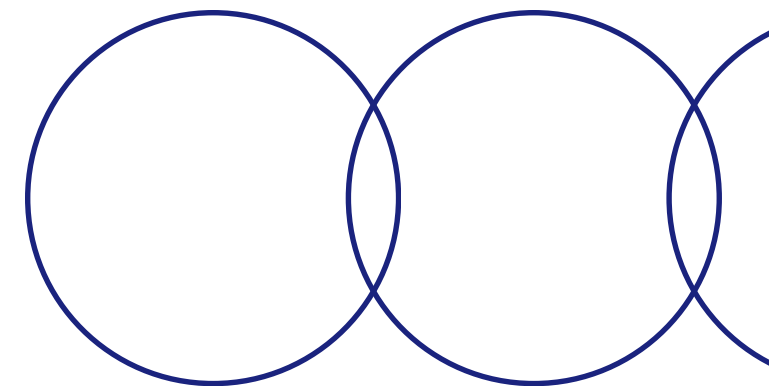
Presented by Jakub Worek and Dawid Zawiański



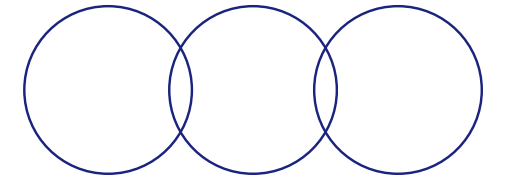
Understanding SLURM Magic

IPython Extension for Commands

SLURM Magic is an **IPython extension** that exposes SLURM commands as magics, allowing users to execute commands like `sbatch`, `squeue`, and `salloc` directly from Jupyter notebook cells.



Project Architecture



Components and Integration

Local Magics Library

The **slurm_magic.py** library provides essential functionalities for SLURM commands, ensuring smooth integration with Jupyter notebooks while incorporating necessary bug fixes for enhanced performance and reliability.

Interactive Notebook

The **slurm_student_lab.ipynb** notebook serves as an interactive exercise, allowing users to learn SLURM functionalities interactively while visualizing results in a user-friendly format within the Jupyter environment.

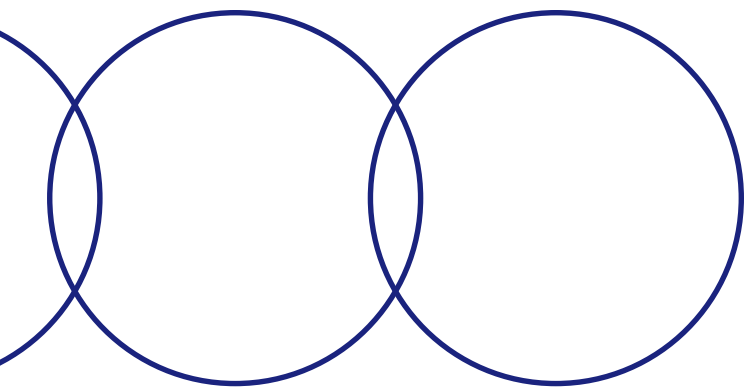
Setup Scripts

The **requirements.txt** and **setup_env.sh** scripts streamline the environment setup process, enabling users to configure their Jupyter lab for optimal SLURM magic usage with minimal effort.

Key Command sbatch

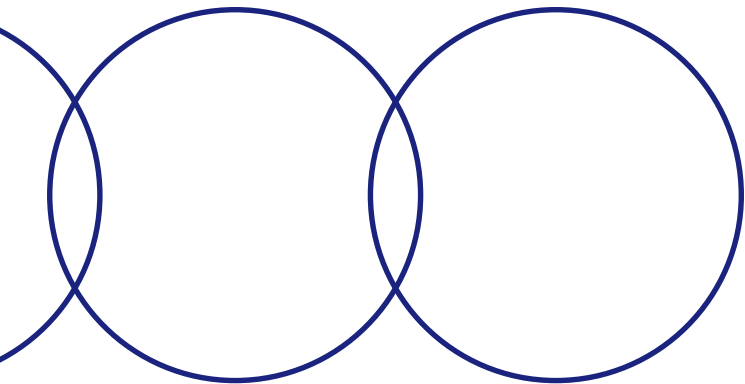
Submitting Batch Jobs

The **sbatch** command allows users to submit cell contents as a batch script to SLURM, returning an assigned Job ID for tracking and management.



Queue Inspection

Understanding job status and cluster information effectively



Job Monitoring

The **squeue** command displays active jobs in the scheduler, allowing users to filter jobs by their username for easy tracking.

Cluster Information

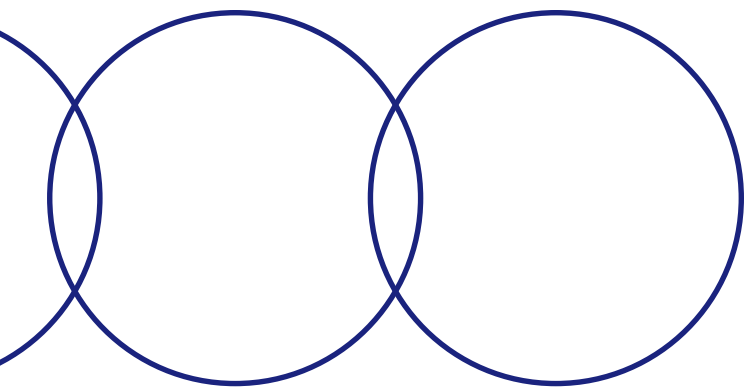
The **sinfo** command provides vital details about cluster nodes and partitions, giving insights into the current state of resources available.

Display Options

Users can view job information in various formats, including raw plain text or as a **pandas DataFrame** for enhanced readability.

Advanced Features

Exploring arrays and dependencies
in SLURM Magic



Job Arrays

Job arrays allow **parallel execution** of tasks, enabling users to submit multiple jobs with a single command for efficiency.

Job Dependencies

Job dependencies facilitate **multi-stage workflows**, ensuring that subsequent jobs execute only after the successful completion of their predecessor jobs.

Partition Limitations

Understanding partition limitations, such as **MaxSubmitPU 1**, is crucial for efficient job submissions, especially when working with testing partitions.

Bug Fixes Overview

Queue Bug

The original library returned a tuple instead of the expected string, causing confusion during job status checks in notebooks.

Output Improvements

Cleaner output is generated without duplicated newlines, ensuring that users can easily read and interpret job status updates.

Code Efficiency

Enhancements to the codebase improve overall performance, ensuring that users experience faster interactions when managing their HPC jobs.

Modal Function

The fixed modal function now correctly handles both string and tuple return types, enhancing compatibility with existing workflows.

Pandas Compatibility

Improved handling of bad lines in data by skipping deprecated error messages, ensuring smoother processing with pandas DataFrames.

User Experience

These bug fixes contribute to a more stable and reliable user experience, making SLURM Magic a valuable tool for Jupyter users.

Student Lab

Learning Modules

The lab features few interactive exercises designed to teach key concepts and workflows related to SLURM and Jupyter integration.

Job Submission

Participants gain hands-on experience submitting **batch jobs** from Jupyter notebooks, enhancing their understanding of HPC job management.

Advanced Patterns

Students will learn about implementing **advanced patterns**, including job arrays and dependencies, to streamline multi-stage computations in HPC.

Cluster Inspection

Students will explore how to **inspect clusters** and monitor jobs effectively using SLURM commands and Jupyter's interactive environment.

Real-time Monitoring

The lab includes exercises on **real-time job monitoring**, using Python to parse output from SLURM's `squeue` command effectively.